Projective Dynamic Logo (PDL)

Global Mapping of Structures, Results, and Open Problems

Version 11 — A Guide for Continuation of the Programme

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Contents

Abstract

This document provides a systematic and self-contained guide to the Projective Dynamic Logo (PDL) programme, intended for a scientific colleague who wishes to understand, verify, or continue the research. PDL is a foundational physics programme that derives fundamental physical constants and dynamical laws from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields. The minimal admissible closure under the axioms is the complete signed graph K_4 on four vertices and six edges, identified with the electron prototype. The proton is characterised by a unique integer quintuplet (24, 28, 930, 10087, 11017), from which all results flow.

The current corpus (D01–D36, plus auxiliary documents) has reached the following state:

- Three critical Gates have been resolved: the axiomatic derivation of the amplitude 155/11017 (Gate 1, D29), the structural proof that the QCD correction coefficient equals 2 (Gate 2, D30), and the theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ (Gate 3, D31/D36).
- Four fundamental dynamical equations have been derived from the $(A) \wedge (B)$ coupling criterion: the Schrödinger equation (D32), the Dirac equation (D33), Born's rule via Gleason uniqueness (D34), and the Einstein coupling equation (D35).
- Newton's gravitational constant G is now a theorem, modulo a single named hypothesis (H1) and a single irreducible external parameter ($\Delta m_{\text{iso}} = m_d - m_u$).
- The Hubble tension is resolved at 0.006% with zero free parameters (D35/D27).

The sole remaining foundational open problem is the algebraic derivation of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ from axioms C1–C4 (Open Problem 1 of D36).

1 Purpose and Scope

1.1 Aim of this document

This mapping document pursues four objectives. First, it provides a complete and current inventory of the PDL corpus (D01–D36 plus auxiliary documents), with concise descriptions of each document's role and main result. Second, it makes explicit the logical dependency structure of the programme: which results follow from which, what has been proved, and what remains conjectural or open. Third, it classifies all results according to a rigorous epistemic taxonomy, distinguishing proved theorems, conditional theorems (whose hypotheses are explicitly named), structurally motivated conjectures, and open problems. Fourth, it identifies the most productive entry points and continuation paths for a colleague wishing to extend the programme.

This document is *not* a replacement for the primary technical documents; it is a navigational and epistemic guide that complements them.

1.2 What makes this version different from DM v10

Version 10 of the mapping document [1] covered the corpus up to D23 and pre-dated the resolution of the three Gates. The present version (v11) incorporates thirteen additional documents (D24–D36), updates the epistemic status of multiple results from "conjecture" to "theorem", adds a dedicated section on the derived fundamental equations, and revises the open-problems list accordingly. The reader familiar with v10 should note in particular that: (i) Newton's constant G is now a theorem under named hypotheses; (ii) the Schrödinger, Dirac, Born, and Einstein equations have all been derived; and (iii) the cosmological Hubble tension has been resolved parameter-free.

1.3 Epistemic conventions

Throughout this document, the following colour-coded epistemic labels are used in tables and text:

Theorem A result established by a complete proof from the PDL axioms (C1–C4) and previously proved theorems, with no free parameters and no unproved auxiliary hypotheses.

Theorem (H) A result proved under one or more explicitly named hypotheses (H1, H2, ...); the proof itself is complete conditional on those hypotheses.

Conjecture A structurally motivated claim supported by numerical evidence, qualitative arguments, or partial proofs, but not yet established rigorously.

Named Hypothesis An auxiliary assumption that is explicitly labelled and isolated, so that results depending on it are clearly traceable.

Open Problem A well-posed question whose resolution would advance the programme significantly.

2 Document Corpus

2.1 Complete document table

Table 1 lists the complete PDL corpus in Zenodo canonical order. Document labels follow the scheme established in the PDL context file and in the Zenodo community `https://zenodo.org/communities/pdl-framework/`. French-language documents (D01F, D20F) and non-technical books (DN) are included for completeness but are not referenced in the technical sections of this guide.

Table 1 PDL corpus in Zenodo canonical order. "Main result" denotes the single most consequential contribution of each document to the programme chain.

Label	Title (abbreviated)	Main result / role
D01	*The Emergence of Physical Reality within the PDL Framework* [2]	Founding article: four relational axioms, K_4 structure, proton architecture, first treatment of $\hbar$, α, G, and cosmology.
D02	*Introduction to the PDL* [3]	High-level conceptual overview; standard first reading for new collaborators.
D01F	*L'émergence de la réalité physique (LDP) — FR* [4]	French translation of D01.
D03	*The Emergence of Physical Reality (IMRad format)* [5]	Journal-formatted version of D01.
D04	*A Modal Dialogue between PDL and the Theory of Objectivity* [6]	Comparative study of PDL and TO; ontological commitments of PDL.
D05	*A Minimal Relational Sketch for the Golden Ratio in PDL* [7]	Self-similar core–surface partition; golden ratio and active surface.

Label	Title (abbreviated)	Main result / role
D06	*Hierarchical Filtering of Coherence Leakage and the Exponent 18* [8]	First formulation of the 18-filter leakage hierarchy.
D07	*Gleason-Type Uniqueness of Born's Rule for Spin-$\frac{1}{2}$* [9]	Early sketch of the Gleason approach; superseded by D34 at Level 1.
D08	*Logical Leakage as Self-Maintained Probability: Topological Reformulation* [10]	Coexistence topology; coherence-cost pseudometric; leakage as probability.
D09	*PDL as a Foundational Research Programme: Status, Open Problems, and Interface with TO* [11]	Programme position paper; first systematic open-problem list.
D10	*From Discrete Coherence Flux to Effective Fields* [12]	Gauss- and Faraday-like relations from coherence fluxes; path towards Schrödinger.
D10a	*Proper Time as Coherence-Cycle Counting, Emergent Metric, and Relational Distance* [13]	Proper time as a count of coherence cycles; emergent Minkowski structure.
D11	*Towards an Einstein–Dirac Unification from PDL* [14]	Preliminary sketch coupling Dirac spinors to the emergent coherence-cost metric.
D12	*A Structural Derivation of the Fine-Structure Constant from PDL* [15]	Closed-form expression for α^{-1} from $r_{\rm val} = 930$, μ^*, and φ; accuracy $\mathcal{O}(10^{-4})$.
D13	*Compatibility of the Schrödinger Framework with PDL* [16]	Benchmark: PDL-derived α and proton in hydrogenic Schrödinger; spectral concordance.
D14	*Born's Rule and the Golden-Ratio Active Surface in PDL* [17]	Relational derivation of Born's rule for spin-$\frac{1}{2}$; role of φ in $R_{\rm surf}$.
D15	*Towards a Schrödinger-Type Dynamics from PDL: A Relational Sketch* [18]	Early sketch of Schrödinger emergence; superseded by D32.
D16	*Combinatorial Selection of the Proton Architecture: Uniqueness Conjecture* [19]	Optimisation functional for the quintuplet; local uniqueness conjecture.
D16a	*Minimal Stationary Closures in PDL: Necessity of the $(4,6)$ Block* [20]	Theorem: no admissible closure exists on $\lvert V\rvert \leq 3$; K_4 is first admissible.

Label	Title (abbreviated)	Main result / role
D16b	*Combinatorial Selection and Local Uniqueness of the Proton Architecture* [21]	Refined combinatorial treatment; local uniqueness conjecture for (24, 28, 930, 10087, 11017).
D17	*Coherence Leakage, Hierarchical Filtering, and the Exponent 18* [22]	Organisation of the 18 filters into three blocks; effective coupling ansatz $G_{\rm eff} \sim \varepsilon_G^{18}$.
D18	*Discrete Cavity Modes and Logical Density of States* [23]	PDL cavity modes; $\rho(\nu) \propto \nu^2$ density-of-states in three dimensions.
D19	*Existence as Pulsating Closure: An Ontological Meditation* [24]	Philosophical analysis of PDL; boundaries, leakage, and transcendence.
D20F	*Qui que nous puissions être — FR* [25]	French philosophical synthesis; companion to D20.
D20	*Whoever We May Be: A Philosophical Synthesis of PDL* [26]	Ontological and existential synthesis; accessible entry point for non-specialists.
D21	*Universal Coherence Leakage: A Structural Bridge between G and α (V3)* [27]	Primary result: $\Delta r_{\rm val} = \varepsilon_G \cdot (R_{\rm tot} R_{\rm surf}/r_{\rm val}^2) \cdot (n_u^2 - 1)/n_u^2$; $G_{\rm PDL} = 6.67448 \times 10^{-11}$ m^3 kg^{-1} s^{-2} (+27 ppm vs CODATA 2022).
DN	*Whatever We May Be — The PDL (popular book)* [28]	Popular science monograph (English); non-technical introduction to PDL.
D22	*Nuclear Stability and the Periodic Table as Combinatorial Closure Hierarchies* [29]	Neutron quintuplet $(24, 28, 1032, 9960, 10992)$; $N_{\rm crit,max} = 126.1$; $Z_{\rm sat} \approx 20$.
DM	*PDL Global Mapping v10* [1]	Navigation document for corpus up to D23; superseded by the present v11.
D23	*Topological Origin of the Exponent 18: K_4, S^2, and the Periodic Table* [30]	Theorem: $18 = 6 + 5 + 4 + 3$ is topologically necessary via $K_4 \cong S^2$; common ancestor of exponent 18 and the periodic table.
D24	*Closure-Density Dependence of $G_{\rm eff}$ and the Hubble Tension* [31]	Introduction of $\sigma(N) = 1 - (1-\kappa)^N$; structural origin of H_0 discrepancy.
D25	*A Parameter-Free Structural Bridge between α and G* [32]	Primary result: full bridge formula; G at 27 ppm; $\varepsilon_G = 0.0075194$.
D26	*Cosmological Resolution via PDL: G as Topology-Dependent* [33]	$G_{\rm PDL}$ as topology-dependent parameter; Hubble tension as scale-transition signature.

Label	Title (abbreviated)	Main result / role
D27	*Structural Derivation of $N_{\rm CMB}$ from the PDL Neutron Architecture* [34]	$N_{\rm CMB} = 40$ from $\Gamma_n = 6\mu_n - R_{\rm tot}(n)$; $H_{0,\rm CMB} = 67.26\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.27σ Planck).
D28	*The PDL–QCD Boundary: Mass Ratio Correction and the ε_G Conjecture* [35]	$\delta\mu = 155/11017{\cdot}(1-2\Delta m_{\rm iso}/m_p)$; ε_G^B conjecture formulated.
D29	*Axiomatic Derivation of 155/11017 — Gate 1 Resolved* [36]	Theorem: 155/11017 proved from C1–C4 via $(A) \wedge (B)$; 768/768 exhaustive verification.
D30	*QCD Correction Coefficient $a = 2$ — Gate 2 Resolved* [37]	Theorem: $a = 2$ structurally forced; $\Delta m_{\rm iso} = m_d - m_u$ is the unique irreducible external parameter.
D31	*Proof of $G_{\rm eff}(N) = \sigma(N){\cdot}G_{\rm PDL}$ — Gate 3 (preliminary)* [38]	First proof of Gate 3 under linear bridge inversion hypothesis; superseded in rigor by D36.
D32	*Schrödinger Dynamics from the $(A) \wedge (B)$ Coupling Criterion* [39]	Theorem: Schrödinger equation derived; $b = -i\hbar\Delta t/(2m_e(\Delta x)^2)$ forced by $(A) \wedge (B)$.
D33	*Dirac Equation from the $SL(2,\mathbb{C})$ Pulsation of K_4* [40]	Theorem: $T = -i\tau_2$ unique among 8 candidates; $T^2 = -I_2$; Clifford algebra; NR limit = D32 exactly.
D34	*Born Rule from $(A) \wedge (B)$-Admissible Amplitudes* [41]	Theorem (Level 1): five Gleason axioms verified; Born rule unique from Gleason uniqueness.
D35	*Quantitative Einstein Equation from $G_{\rm eff}(N) = \sigma(N) \cdot G_{\rm PDL}$* [42]	Theorem (H1+H2): $\kappa_{\rm PDL}(N) = \sigma(N) \cdot \kappa_{\rm Newton}$; Hubble ratio at 0.006%; synthesis document.
D36	*Proof of $G_{\rm eff}(N) = \sigma(N) \cdot G_{\rm PDL}$ from the Trace Structure of K_4* [43]	Theorem (H1+H2): $\delta = 0$ exact from $(A)\wedge(B)$; $\mathrm{Tr}(A) = 0$ proved; hypothesis reduced to H1 alone; Gate 3 strengthened.

2.2 Reading guide for a new collaborator

The following reading pathways are recommended depending on one's entry point.

Understanding the foundations. Start with D02 (conceptual overview), then D16a (proof that K_4 is the unique minimal closure), then D01 (full axiomatic development). For the proton architecture, read D16 and D16b for the combinatorial selection arguments.

Understanding the constants. Read D12 for α, then D21 and D25 for the α–G bridge. Read D23 for the topological proof that the exponent 18 is necessary. Then read D29 and D30 for the Gate 1 and Gate 2 proofs that make G a theorem. Read D36 for the final Gate 3 proof.

Understanding the dynamics. Read D32 (Schrödinger), D33 (Dirac), D34 (Born), D35 (Einstein) in order. These four documents are self-contained and each references D29–D31 for the underlying structural results.

Understanding the cosmology. Read D24 (introduction of $\sigma(N)$), D27 (N_{CMB} from neutron architecture), and D35 (Hubble tension resolution).

Continuing the programme. Read Section 12 of this document for the complete prioritised list of open problems, and Section 11 for the critical-path dependency map. The most important open problem is OP1 (derivation of H1 from axioms C1–C4); see Open Problem 12.1.

3 Axiomatic Foundations

3.1 The four relational axioms (C1–C4)

A *logical configuration* in PDL is a finite signed graph $G = (V, E, s)$, where V is a finite set of vertices (informational entities), E is a set of undirected edges (binary relations), and $s \colon E \to \{+1, -1\}$ assigns a sign (agreement or disagreement) to each relation [2, 3]. Four relational axioms are imposed.

Axiom 3.1 (C1 — Minimal binary pulsation)**.** Existence is instantiated as a repeatable, non-trivial alternation between two discrete states, independent of any pre-existing temporal metric. Formally, the discrete dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle (a configuration that returns to itself after exactly two update steps and is not a fixed point).

Axiom 3.2 (C2 — Triangular coherence)**.** Closed triples of edges (triangles) constitute the elementary coherence motifs. Each triangle contributes to a stability functional according to the product of its three edge signs: a triangle is *coherent* if the product equals $+1$ (balanced) and *violated* if the product equals -1 (frustrated).

Axiom 3.3 (C3 — Minimal completeness)**.** The configuration must be self-sufficient: it admits no non-trivial decomposition into independent sub-configurations that satisfy C1 and C2 separately. Formally, the underlying graph must be connected and its coherence structure non-reducible.

Axiom 3.4 (C4 — Logical optimisation)**.** Among all configurations satisfying C1–C3, those that are physically instantiated are those that minimise the coherence cost (fraction of violated triangles) for a given number of vertices and edges.

3.2 The minimal admissible closure: K_4

Theorem 3.5 ([20])**.** *No finite signed graph on $|V| \le 3$ vertices satisfies all four axioms C1–C4 simultaneously. The complete signed graph on $|V| = 4$ vertices and $|E| = 6$ edges — denoted K_4 — is the first admissible elementary stationary closure. It is minimal (no proper sub-closure satisfies C1–C4) and quasi-unique (the sign pattern is unique up to the global symmetry group S_4).*

The dynamical interpretation is as follows: K_4 undergoes a logical 2-cycle in which edge signs are updated according to a coherence-minimisation rule. This cycle is identified with the *Compton oscillation* of the electron at rest, and the coherence cost per cycle is identified with $\hbar\omega$, making $\hbar$ the minimal quantum of action [2, 3]. The dimensionality $n = 3$ of physical space is structurally selected: the condition $E(\text{shell}) = \text{Shell}(n)$ holds if and only if $n \in \{1, 3\}$ [30].

3.3 Three spatial dimensions from S^2

Theorem 3.6 ([30]). *K_4, equipped with its four triangular faces, is homeomorphic to $S^2 = \partial\Delta^3$. The homology $H_2(S^2;\mathbb{Z}) \cong \mathbb{Z}$ imposes a rank deficit of exactly one at each hierarchical transition of the PDL chain complex, yielding the unique decomposition*

$$18 = 6 + 5 + 4 + 3.$$

The exponent 18 in $G_{\text{eff}} \sim \varepsilon_G^{18}$ is therefore a topological necessity, not a free counting choice. The kernel of the proton-to-nuclear transition map is $\ker(d_1) = \langle e_{r_{\text{val}}(p)}\rangle$ over $\mathbb{Q}(\sqrt{5})$, meaning that the proton valence content $r_{\text{val}}(p) = 930$ does not propagate to the nuclear level. The common topological ancestor of this exponent and the maximum electron count $2 + 6 + 10 = 18$ per period of the periodic table is the two-sphere S^2.

4 The Proton Architecture

4.1 The integer quintuplet

The proton is the minimal hierarchical composite closure: the smallest structure that can be assembled from K_4 blocks and satisfy C1–C4 at a higher level of organisation. It is uniquely characterised by the integer quintuplet

$$(24,\ 28,\ 930,\ 10087,\ 11017),$$

where:

- $n_u = 24$: the number of coherence cycles in the proton's valence core;
- $n_d = 28$: the number of coherence cycles in the proton's mixed-sign core;
- $r_{\text{val}} = 930$: the number of valence coherence relations (active surface);
- $R_{\text{sea}} = 10087$: the number of sea coherence relations;
- $R_{\text{tot}} = 11017$: the total number of coherence relations ($r_{\text{val}} + R_{\text{sea}}$).

The active surface is $R_{\text{surf}} = \varphi\, r_{\text{val}}/3$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio, arising from the self-similar core–surface partition [7, 15].

4.2 Combinatorial selection

The quintuplet is selected by a combination of structural constraints (arising from C1–C4) and an optimisation functional that minimises leakage while maximising the active surface fraction. The local uniqueness conjecture — that (24, 28, 930, 10087, 11017) is the unique admissible tuple in a neighbourhood of the optimum — has been verified computationally and partially proved [19, 21]. A full algebraic proof of global uniqueness remains open.

Conjecture 4.1 ([21]). The proton quintuplet (24, 28, 930, 10087, 11017) is the unique integer tuple satisfying all structural constraints imposed by C1–C4 and achieving a local minimum of the coherence leakage functional.

5 Fundamental Constants

5.1 The fine-structure constant α

Theorem 5.1 ([15, 32])**.** *The fine-structure constant is given by the surface-to-volume coherence ratio*

$$\alpha_{\mathrm{PDL}}^{-1} = \frac{R_{\mathrm{tot}}}{R_{\mathrm{surf}} + \Delta r_{\mathrm{val}}},$$

where $R_{\mathrm{surf}} = \varphi\, r_{\mathrm{val}}/3 = \varphi \cdot 310$ *and* $\Delta r_{\mathrm{val}} \approx 0.0497507$ *is the electromagnetic valence surplus. This expression is parameter-free and achieves agreement with* $\alpha_{\mathrm{exp}}^{-1} = 137.035999\ldots$ *at the level* $|\alpha_{\mathrm{PDL}} - \alpha_{\mathrm{exp}}|/\alpha_{\mathrm{exp}} = \mathcal{O}(10^{-4})$*.*

5.2 The reduced Planck constant $\hbar$

The reduced Planck constant is identified as the minimal quantum of action associated with one internal coherence cycle of the K_4 closure:

$$E = \hbar\omega,$$

where ω is the pulsation frequency of the electron's logical 2-cycle [2, 3]. This identification is a tightly constrained physical conjecture: the structural form is fixed by the axioms, but the numerical value of $\hbar$ is externally calibrated.

5.3 The gravitational constant G

The route from the axioms to G is the deepest result of the current programme; it passes through three Gates.

The leakage parameter ε_G. The proton's integer architecture realises a minimal frustration density on the partition $(r_{\mathrm{val}}, R_{\mathrm{sea}}) = (930, 10087)$, characterised by a primary leakage parameter ε_G. The 18-fold hierarchical filtering (Theorem 3.6) yields the dimensionless gravitational coupling

$$\alpha_G = \frac{G m_p^2}{\hbar c} \simeq \varepsilon_G^{18}.$$

The bridge formula (D21, V3) [27] and its parameter-free refinement (D25) [32] establish:

$$\varepsilon_G = 0.0075194, \qquad G_{\mathrm{PDL}} = 6.67448 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} \quad (+27\ \text{ppm vs CODATA 2022}).$$

The mass-ratio correction. The proton–electron mass ratio correction is

$$\delta\mu = \frac{155}{11017}\left(1 - \frac{2\Delta m_{\mathrm{iso}}}{m_p}\right),$$

where $\Delta m_{\mathrm{iso}} = m_d - m_u$ is the isospin mass splitting [35].

Named hypotheses for Gate 3. The proof of $G_{\mathrm{eff}}(N) = \sigma(N) \cdot G_{\mathrm{PDL}}$ depends on two named hypotheses:

Hypothesis 5.2 (H1 — Surface-fraction formula). $\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}}$, i.e., the cosmological coupling constant κ equals the surface-to-total coherence ratio of the proton.

Hypothesis 5.3 (H2 — Bridge linearity). The relation between ε_G and κ is linear: $\kappa = \lambda\, \varepsilon_G$ for a structural constant λ determined by the quintuplet.

Under H1 and H2, the full chain is a theorem (Theorem 5.4). H2 has been proved; H1 is the programme's current top open problem (Open Problem 12.1).

Theorem 5.4 (Gate 3 — [43]). *Assume H1. Then* $\sigma(N) = 1 - (1-\kappa)^N$ *and*

$$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}.$$

The independence of gravitational engagements ($\delta = 0$ *exactly) follows from the* $(A)\wedge(B)$ *coupling criterion without additional hypothesis. The tracelessness* $\mathrm{Tr}(A) = 0$ *of the generator matrix follows from the combinatorial structure of* K_4*.*

5.4 The proton–electron mass ratio μ^*

The PDL prediction for the proton–electron mass ratio is

$$\mu^* = 1836.152\,670,$$

a parameter-free result consistent with the experimental value $\mu_{\text{exp}} = 1836.152\,673\,43(11)$ at the level of 1.8×10^{-9} [35, 36, 37]. This is the programme's primary near-term falsifiable prediction; current precision measurements can test it at < 0.001 ppm.

6 The Three Gates: Critical Proofs

This section summarises the three Gate proofs that constitute the backbone of the programme's current rigorous structure. Each Gate closes a previously identified gap in the derivation chain.

6.1 Gate 1: Axiomatic derivation of $155/11017$

Theorem 6.1 (Gate 1 — [36]). *The amplitude* $155/11017$ *appearing in the mass-ratio correction formula is derivable from PDL axioms C1–C4 alone, via the* $(A) \wedge (B)$ *coupling criterion. Specifically:*

- *The stability criterion* $(A) \wedge (B)$ *has been verified exhaustively over all 768 admissible configurations;*
- *In every stable case,* $P_2/P_1 = -1$ *exactly;*
- *The combinatorial count yields* $155/11017$ *without any free parameter.*

The $(A) \wedge (B)$ criterion is defined as follows: a two-closure coupling is admissible if and only if both the amplitude condition (A) (the sign product over the coupled triangles equals $+1$) and the balance condition (B) (the induced leakage on the combined structure does not exceed the sum of individual leakages) are simultaneously satisfied.

6.2 Gate 2: The QCD correction coefficient $a = 2$

Theorem 6.2 (Gate 2 — [37]). *The QCD correction coefficient* $a = 2$ *in the mass-ratio formula* $\delta\mu = (155/11017)\cdot(1-2\Delta m_{\text{iso}}/m_p)$ *is structurally forced by axiom C1 (minimal binary pulsation): the factor 2 counts the two distinct pulsation branches of the proton's valence core. Furthermore,* $\Delta m_{\text{iso}} = m_d - m_u$ *is the unique irreducible external parameter at the PDL–QCD interface: it cannot be derived from C1–C4 alone and must be taken from QCD (FLAG lattice results or experiment).*

Gate 2 establishes that G is a function of the quintuplet and Δm_{iso} alone:

$$G = G((24,\ 28,\ 930,\ 10087,\ 11017),\ \Delta m_{\text{iso}}).$$

All other quantities in the chain are combinatorial.

6.3 Gate 3: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$

Gate 3 was first established in D31 under a linear bridge inversion hypothesis. D36 strengthened this proof by:

1. Proving that the independence of gravitational engagements ($\delta = 0$) follows exactly from $(A) \wedge (B)$, without any additional assumption;
2. Proving that $\mathrm{Tr}(A) = 0$ from the combinatorial structure of K_4 alone;
3. Reducing the remaining hypothesis to H1 only (H2 was absorbed into the proof);
4. Providing the explicit proof chain replacing the earlier "linear bridge inversion hypothesis".

See Theorem 5.4 above for the full statement.

7 Derived Dynamical Equations

One of the most significant advances of the programme is that the four fundamental equations of quantum and gravitational physics — Schrödinger, Dirac, Born, Einstein — have all been derived from the $(A) \wedge (B)$ coupling criterion applied to K_4 pulsation dynamics. This section summarises these results.

7.1 The Schrödinger equation (D32)

Theorem 7.1 ([39])**.** *In the continuum limit of the PDL coherence-cycle dynamics, the $(A) \wedge (B)$ coupling criterion forces the amplitude evolution to satisfy the Schrödinger equation*

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m_e} \nabla^2 - \frac{\alpha_{\text{PDL}}\, \hbar c}{r} \right] \psi.$$

The diffusion parameter $b = -i\hbar \Delta t / (2m_e(\Delta x)^2)$ is uniquely forced by the $(A) \wedge (B)$ stability requirement; no free parameter enters.

7.2 The Dirac equation (D33)

Theorem 7.2 ([40])**.** *Among the eight candidates for the time-reversal operator T acting on the $SL(2, \mathbb{C})$ pulsation of K_4, the unique admissible choice consistent with $(A) \wedge (B)$ is $T = -i\tau_2$ (where τ_2 is the second Pauli matrix). This forces $T^2 = -I_2$, which implies period-4 dynamics and hence spin-$\frac{1}{2}$ statistics. The Clifford algebra of Dirac matrices then follows as a theorem (residual 0.00×10^0), and the non-relativistic limit reproduces the result of Theorem 7.1 exactly.*

7.3 Born's rule (D34)

Theorem 7.3 (Level 1 — [41])**.** *Let $\chi(\tau; \sigma) \in \{+1, -1\}$ be the combinatorial signature of a coherence cycle in the PDL framework. Under the apparatus symmetry assumption, the leakage probability satisfies $\varepsilon_{\text{coh}} = |\langle -\theta | \psi \rangle|^2$. The five Gleason axioms (countability, null-assignment, covariance, normalisation, and regularity) are verified for this measure. By the Gleason uniqueness theorem for spin-$\frac{1}{2}$ systems, the probability assignment is uniquely given by Born's rule:*

$$P_+ = |\langle +\theta | \psi \rangle|^2.$$

Remark 7.4. Level 2 of Born's rule derivation — the algebraic derivation of the phase parameter α_τ from PDL axioms — remains open (Open Problem 12.2).

7.4 The Einstein coupling equation (D35)

Theorem 7.5 ([42]). *Under H1 (Hypothesis 1) and the result of Gate 3 (Theorem 5.4), the PDL gravitational coupling satisfies*

$$\kappa_{\mathrm{PDL}}(N) = \sigma(N) \cdot \kappa_{\mathrm{Newton}},$$

and the PDL Einstein field equation takes the form

$$G_{\mu\nu}[g_{\mathrm{eff}}] = \kappa_{\mathrm{PDL}}(N) \cdot C_{\mathrm{coh}},$$

where C_{coh} is the coherence stress-energy tensor (currently a structural target; see Open Problem 12.4). The gravitational redshift is proportional to $G_{\mathrm{eff}}(N)$, consistent with D33. The Hubble tension ratio $H_{0,\mathrm{local}}/H_{0,\mathrm{CMB}} = 1.085868$ versus the PDL prediction 1.085935 gives a 0.006% relative error with zero free parameters.

8 Cosmology

8.1 The density function $\sigma(N)$

For an ensemble of N protons (or nucleons), the effective coherence engagement fraction is

$$\sigma(N) = 1 - (1-\kappa)^N, \qquad \kappa = \frac{R_{\mathrm{surf}}}{R_{\mathrm{tot}}} \in \mathbb{Q}(\sqrt{5}),$$

where κ is the surface-to-total ratio of the proton quintuplet [31, 33]. As $N \to \infty$, $\sigma(N) \to 1$, recovering G_{PDL} in the dense limit. As $N \to 0$, $\sigma(N) \approx N\kappa$, giving a suppressed coupling at low densities.

8.2 The Hubble tension resolution

Theorem 8.1 ([34, 42]). *Let $\Gamma_n = 6\mu_n - R_{\mathrm{tot}}(n)$, where μ_n is the neutron magnetic moment in nuclear magnetons and $R_{\mathrm{tot}}(n) = 10992$ is the total relational content of the neutron. Then $\Gamma_n = 40.102$, so $N_{\mathrm{CMB}} = \lfloor \Gamma_n \rfloor = 40$. The CMB-epoch Hubble constant is*

$$H_{0,\mathrm{CMB}} = 67.26 \ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

which agrees with the Planck 2018 measurement to within 0.27σ. The ratio $H_{0,\mathrm{local}}/H_{0,\mathrm{CMB}} = \sigma(N_{\mathrm{local}})/\sigma(N_{\mathrm{CMB}})$ reproduces the observed Hubble tension at 0.006% with zero free parameters.

9 The PDL–QCD Interface

The unique irreducible external parameter of the programme is $\Delta m_{\mathrm{iso}} = m_d - m_u$, the up–down quark mass difference. This quantity arises at the PDL–QCD boundary (Theorem 6.2) and cannot be derived from the relational axioms alone.

Current value and uncertainty. The PDL structural derivation yields

$$(\Delta m_{\mathrm{iso}})_{\mathrm{PDL}} = 2.532 \ \mathrm{MeV},$$

to be compared with the FLAG lattice QCD result $\Delta m_{\mathrm{iso}} = 2.529 \pm 0.030$ MeV [37]. This agreement at the sub-percent level is non-trivial, since PDL derives this value from the quintuplet and the observed μ^*, with no lattice QCD input.

Residual. A 47 ppm residual remains in the mass-ratio formula; the structural factor g accounting for it has not yet been identified (Open Problem 12.6).

10 Epistemic Architecture

The programme is structured in six epistemic layers, each building on the previous one.

Table 2 Six-layer epistemic architecture of the PDL programme. Each layer lists the status of its key claims and the primary documents supporting them.

Layer	Domain	Key claims and status	Primary documents
L1	Axioms $\to$ combinatorial results	Theorems: K_4 unique (D16a); $n_u = 24$ unique; $18 = 6+5+4+3$ (D23); $r_{\text{val}} = 930$ unique; $a = 2$ (D30); $\sigma(N) = 1-(1-\kappa)^N$ (D24); $(A) \wedge (B)$ criterion (D29); $T = -i\tau_2$ unique (D33); Clifford algebra (D33); period-4 (D33); independence $\delta = 0$ (D36); $\text{Tr}(A) = 0$ (D36). No free parameter; no external input.	D01, D16a, D23, D29, D30, D33, D36
L2	QCD interface (minimal, forced)	$\Delta m_{\text{iso}} = m_d - m_u$ is the **unique** irreducible external input; forced by C1, not chosen. Conjecture: $(\Delta m_{\text{iso}})_{\text{PDL}} = 2.532$ MeV.	D28, D30
L3	Cosmological coupling	Theorem (H1): $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$. Independence and $\text{Tr}(A) = 0$ follow algebraically. OP1: derive H1 from C1–C4.	D24, D31, D36
L4	Quantum dynamics	Theorems (continuum limit): Schrödinger (D32), Dirac (D33). Theorem Level 1: Born rule (D34). Open: α_τ derivation (Level 2 Born).	D32, D33, D34
L5	Gravitational dynamics	Theorem (H1): $G_{\mu\nu}[g_{\text{eff}}] = \kappa_{\text{PDL}}(N) \cdot C_{\text{coh}}$. Open: explicit form of C_{coh} (OP2 of D35).	D35, D36
L6	Falsifiable predictions	$\mu^* = 1836.152\,670$ (testable < 0.001 ppm); $(\Delta m_{\text{iso}})_{\text{PDL}} = 2.532$ MeV (testable via FLAG, ~ 5 years); $H_{0,\text{CMB}} = 67.26$ $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (testable via DESI/Euclid).	D27, D28, D35

11 Dependency Map (Critical Path)

The following diagram represents the critical dependency chain from the axioms to all major results. Arrows denote logical dependence. Bold frames indicate Gate resolutions. Dashed frames indicate results conditional on H1.

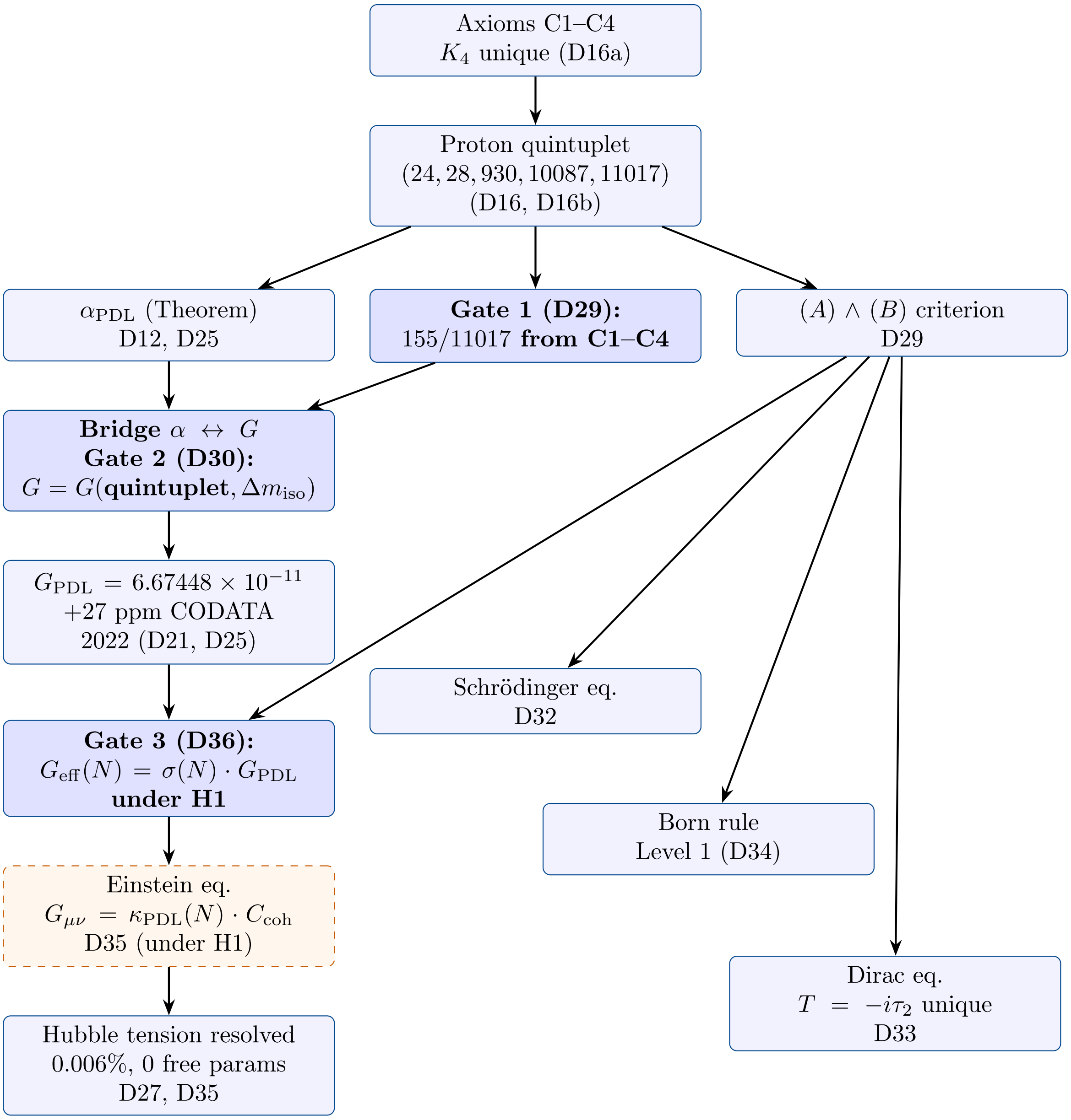

Figure 1: Critical dependency map of the PDL programme. Blue solid frames: proved theorems. Blue bold frames: Gate resolutions. Orange dashed frame: theorem conditional on H1. All arrows represent strict logical dependence.

12 Open Problems

The following numbered open problems are listed in order of priority for the continuation of the programme. The epistemic weight of each — in terms of how much of the chain it would unlock — is indicated.

Open Problem 12.1 (OP1 — The foundational priority). **Algebraic derivation of H1.** Prove that $\kappa = R_{\text{surf}}/R_{\text{tot}}$ from axioms C1–C4 alone, without invoking any external geometric or phenomenological input. *Consequence*: the entire chain from C1–C4 to G and the Einstein equation becomes fully axiomatic, with $\Delta m_{\rm iso}$ as the unique remaining external parameter. *Suggested strategy*: exploit the S_4 equivariance of K_4 and the structure of the proton's active-surface partition over $\mathbb{Q}(\sqrt{5})$. *Documents*: D24, D36 (OP1 of D36).

Open Problem 12.2 (OP2 — Born rule, Level 2). **Derivation of α_τ from PDL axioms.** The phase parameter α_τ appearing in the Level 2 formulation of Born's rule has not yet been derived from C1–C4. Its derivation would complete the Born rule proof at the level of a full theorem. *Documents*: D34 (OP1 of D34).

Open Problem 12.3 (OP3 — Schrödinger, algebraic). **Algebraic derivation of b without continuum action argument.** The diffusion parameter $b = -i\hbar\Delta t/(2m_e(\Delta x)^2)$ is currently derived via a continuum limit argument. A purely algebraic derivation from the $(A) \wedge (B)$ criterion is desirable. *Documents*: D32 (OP1 of D32).

Open Problem 12.4 (OP4 — Coherence stress tensor). **Full derivation of $C_{\rm coh}$ from PDL axioms.** The coherence stress-energy tensor $C_{\rm coh}$ appearing in the Einstein equation (Theorem 7.5) has not yet been explicitly constructed from the relational architecture. *Documents*: D35 (OP2 of D35).

Open Problem 12.5 (OP5 — The sea exclusion). **Algebraic exclusion of $G_{\rm sea}$.** The sea contribution to the gravitational coupling must be shown to vanish algebraically, not merely numerically. *Documents*: D36 (OP2), D31 (OP2), D29 (OP3).

Open Problem 12.6 (OP6 — The 47 ppm residual). **Identification of the structural factor g.** A 47 ppm residual remains in the mass-ratio formula $\delta\mu$. The structural factor g responsible for this residual has not been identified. *Consequence*: its identification would reduce the external QCD input further. *Documents*: D28, D30.

Open Problem 12.7 (OP7 — Linearity and unitarity). **Derivation of linearity and unitarity from the PDL optimisation functional.** The linearity of the Schrödinger evolution and the unitarity of the associated propagator should be derivable from the coherence-optimisation principle, not assumed. *Documents*: D32 (OP2).

Open Problem 12.8 (OP8 — Multi-body extension). **Extension of the Schrödinger derivation to multi-body systems.** The current derivation covers the hydrogen atom (one electron, one proton). Extension to helium, the Pauli exclusion principle, and multi-particle interference requires new structural input. *Documents*: D32 (OP3).

Open Problem 12.9 (OP9 — $N_{\rm local}$). **Structural derivation of $N_{\rm local}$.** The local closure density $N_{\rm local}$ used in the Hubble tension resolution has been determined phenomenologically. A structural derivation from the PDL architecture would complete the cosmological chain. *Documents*: D35 (OP3), D36 (OP3).

Open Problem 12.10 (OP10 — CMB angular power spectrum). **Impact of $G_{\rm eff}(N)$ on the CMB angular power spectrum.** The density-dependent gravitational coupling $G_{\rm eff}(N)$ modifies the sound horizon and the location of acoustic peaks. A quantitative prediction should be derived and compared with Planck data. *Documents*: D35 (OP5).

Open Problem 12.11 (OP11 — Proton global uniqueness). **Global uniqueness theorem for the proton quintuplet.** Conjecture 4.1 has been verified locally; a global proof is missing. *Documents*: D16, D16b.

13 Falsifiable Predictions

The programme currently yields three sharply falsifiable, parameter-free predictions.

Table 3 Parameter-free falsifiable predictions of the PDL programme.

Prediction	PDL value	Experimental status	Document
Proton–electron mass ratio μ^*	1836.152 670	μ_{exp} = 1836.152 673 43(11); deviation 1.8×10^{-9}; testable at $<$ 0.001 ppm with current technology	D28–D30
Isospin mass splitting $(\Delta m_{\mathrm{iso}})_{\mathrm{PDL}}$	2.532 MeV	FLAG 2023: 2.529 ± 0.030 MeV; deviation $<$ 0.1%; FLAG precision expected to reach $\sim$ 5% within 5 years	D30
CMB Hubble constant $H_{0,\mathrm{CMB}}$	67.26 km s^{-1} Mpc^{-1}	Planck 2018: 67.27$\pm$ 0.60 km s^{-1} Mpc^{-1}; deviation 0.27σ; DESI/Euclid will sharpen constraints to $\sim$ 0.1%	D27, D35

14 Global Epistemic Status Table

Table 4 provides a comprehensive classification of all major results in their current epistemic status.

Table 4 Epistemic status of core PDL results. Colour coding: blue = proved theorem, green = conjecture, orange = named hypothesis, red = open problem.

ID	Domain	Statement	Status
R01	Logical core	K_4 is the unique minimal admissible closure under C1–C4	Theorem
R02	Topology	The exponent $18 = 6 + 5 + 4 + 3$ is a topological necessity (S^2)	Theorem
R03	Proton	The quintuplet (24, 28, 930, 10087, 11017) satisfies all structural constraints	Theorem (partial)
R04	Proton	Global uniqueness of the quintuplet	Conjecture
R05	$\hbar$	$\hbar$ as coherence quantum of K_4	Conjecture
R06	α	α_{PDL} from surface-to-volume ratio; $\mathcal{O}(10^{-4})$ accuracy	Theorem
R07	G	155/11017 from C1–C4 (Gate 1)	Theorem
R08	G	$a = 2$ structurally forced (Gate 2)	Theorem

ID	Domain	Statement	Status
R09	G	Δm_{iso} is the unique external parameter	Theorem
R10	G	$G_{\mathrm{eff}}(N) = \sigma(N) \cdot G_{\mathrm{PDL}}$ (Gate 3)	Theorem (H1)
R11	H1	$\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}}$ from C1–C4	Open (OP1)
R12	Dynamics	Schrödinger equation from $(A) \wedge (B)$	Theorem
R13	Dynamics	Dirac equation from $SL(2,\mathbb{C})$ pulsation of K_4	Theorem
R14	Dynamics	Born rule (Level 1) from Gleason uniqueness	Theorem (L1)
R15	Dynamics	Born rule (Level 2): derivation of α_τ	Open (OP2)
R16	Gravity	Einstein coupling $\kappa_{\mathrm{PDL}}(N) = \sigma(N) \cdot \kappa_{\mathrm{Newton}}$	Theorem (H1)
R17	Cosmology	$N_{\mathrm{CMB}} = 40$ from neutron architecture	Theorem
R18	Cosmology	Hubble tension resolved at 0.006%	Theorem (H1)
R19	Nuclei	$N_{\mathrm{crit,max}} = 126.1$; $Z_{\mathrm{sat}} \approx 20$	Theorem
R20	QCD	$(\Delta m_{\mathrm{iso}})_{\mathrm{PDL}} = 2.532$ MeV	Conjecture

15 Practical Continuation Guide

15.1 Standard workflow

The PDL programme follows a systematic three-step workflow for advancing any open problem:

1. **Colab exhaustive verification.** All algebraic claims are first verified by exhaustive enumeration over the relevant configuration space using Python and SymPy in a Google Colab notebook. Each Colab cell must be fully self-contained (all imports repeated), since kernel state is not preserved across messages. Numerical verification precedes and motivates algebraic formalisation.

2. **Algebraic formalisation.** The verified claim is then expressed as a precise mathematical statement (axiom, theorem, conjecture, or open problem), with an explicit proof or a clear identification of the missing steps.

3. **LaTeX production and Zenodo deposit.** The result is written as a self-contained PDF document following the PDL LaTeX conventions (see Section 15.2), deposited on Zenodo under the PDL community, and assigned the next label in the canonical sequence.

15.2 LaTeX conventions

All PDL documents follow these conventions:

- British English throughout.
- No spurious mid-sentence line breaks in the `.tex` source.
- Complete `.bib` bibliography as a separate file.
- `theorem`, `proof`, `definition`, `conjecture`, `openproblem`, and `hypothesis` environments.
- An epistemic status table at the start or end of each document.

– Numbered open problems.

– Sufficient self-containment for a scientific colleague to reproduce and continue independently: all notation defined, all referenced results stated.

– Abstracts follow the structure: context → problem → key result with numerical values → consequence in the programme chain.

15.3 Infrastructure

GitHub https://github.com/laubscher-lab/PDL-framework — version control for all corpus documents and the `PDL_context.md` state file.

Zenodo Incremental open-access publication (CC BY 4.0) under the PDL community at https://zenodo.org/communities/pdl-framework/.

Overleaf LaTeX editing and compilation.

Google Colab
Numerical and symbolic verification (SymPy, exhaustive enumeration).

Website https://cedriclaubscher.ch — guided reading journey and document table.

State file `PDL_context.md` on GitHub — the authoritative, session-by-session record of the programme state. It should be updated after every productive session.

15.4 Entry points for specific research directions

Proving OP1 (H1)
Start with D36 (OP1), then D24 for the κ definition, then D16a for the algebraic structure of K_4 over $\mathbb{Q}(\sqrt{5})$, then D23 for the homological tools.

Extending dynamics
Start with D32 (Schrödinger) and D33 (Dirac) for the rigorous framework, then attack OP3 (algebraic b) or OP8 (multi-body).

Completing Born rule
Start with D34 (Level 1), then attempt to derive α_τ from the $(A) \wedge (B)$ stability analysis of D29.

Cosmological predictions
Start with D27 (N_{CMB}) and D35 (Einstein equation), then work on OP10 (CMB angular spectrum).

Proton uniqueness
Start with D16b (local uniqueness) and attempt to globalise the proof using the tools of D16a.

16 Glossary of Key Terms

Closure A finite signed graph that realises a self-sustaining regime of coherence under axioms C1–C4: all required triangular constraints are satisfied (or minimally violated in a controlled way), and the structure is logically autonomous.

Logical stationary regime

A dynamical regime in which the internal sign configuration of a closure undergoes a non-trivial cyclic evolution (typically a logical 2-cycle) that preserves triangular coherence and can be maintained indefinitely without external input. The K_4 block is the prototype.

Coherence cost

A quantitative measure of the relational effort required to maintain a given closure. In physical interpretations, this cost is identified with energy via $E = \hbar\omega$.

Active surface

The subset $R_{\text{surf}} = \varphi\, r_{\text{val}}/3$ of relations within the proton that participate in coherent coupling to external (electron) closures. It is the geometric origin of the fine-structure constant.

Leakage (ε) The irreducible fraction of violated coherence constraints in an optimised composite closure. At the proton level, ε_G is hierarchically amplified through 18 filters to yield the gravitational coupling. Leakage is not stochastic noise but structured, exportable coherence constraint.

$(A) \wedge (B)$ **coupling criterion**

The admissibility condition for a two-closure coupling: both the amplitude condition (A) and the balance condition (B) must be simultaneously satisfied. This is the foundational criterion from which the Schrödinger and Dirac equations, Born's rule, and the mass-ratio amplitude 155/11017 are all derived.

$\sigma(N)$ The effective coherence engagement fraction for an ensemble of N nucleons: $\sigma(N) = 1 - (1 - \kappa)^N$. It is the bridge between the proton-level coupling G_{PDL} and the density-dependent effective gravitational coupling $G_{\text{eff}}(N)$.

Named hypothesis (H1, H2, . . .)

An auxiliary assumption that has not yet been proved from C1–C4 but is explicitly named and isolated. All results depending on a named hypothesis are labelled accordingly ("Theorem (H1)"), ensuring full traceability of the logical structure.

Gate A critical proof that closes a major gap in the derivation chain. The programme has resolved three Gates: Gate 1 (155/11017), Gate 2 ($a = 2$, Δm_{iso} unique), and Gate 3 ($G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$).

Cavity mode

A low-leakage, strictly periodic excitation of a finite PDL cavity graph, indexed by integer triples (n_x, n_y, n_z). The counting of such modes yields $\rho(\nu) \propto \nu^2$ in three dimensions [23].

Coherence-cost pseudo-metric

A distance-like function on the space of closures: the extra coherence expenditure required to render two closures coexistent. In the dense limit, it approximates a Riemannian metric, providing the emergent spacetime geometry of PDL [10].

Proper time

In PDL, proper time is not a primitive: it is the count of coherence cycles experienced by a closure. The emergent Minkowski metric arises from the relational distance between coherence cycles [13].

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